

ETHICAL HACKING ANSWERS

Question 1: Map out the network topology

a) What is the correct tool for this job?

The correct tool for mapping network topology is Nmap.

Nmap is used for:

- Host discovery (finding active devices)
- Port scanning
- Service detection
- OS detection
- Network mapping

It operates mainly on:

- OSI Layer 3 (Network layer) → IP discovery
- OSI Layer 4 (Transport layer) → Port scanning

b. Performing a Comprehensive scan of the local network and identified:

My scan network is:

10.15.164.0/24 (24 is the subnet)

i. Host discovery scan performed:

This scan identified active hosts on the 10.15.164.0/24 network including my Kali machine (10.15.164.71) and other reachable devices.

```
(kali@kali)-[~]
$ hostname -I
10.15.164.71

(kali@kali)-[~]
$

(kali@kali)-[~]
$ nmap -sn 10.15.164.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 21:56 -0400
Nmap scan report for 10.15.164.74
Host is up (0.00036s latency).
MAC Address: 64:D6:9A:B1:FB:CA (Intel Corporate)
Nmap scan report for 10.15.164.85
Host is up (0.030s latency).
MAC Address: A6:FF:1F:5C:09:22 (Unknown)
Nmap scan report for 10.15.164.71
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.87 seconds

(kali@kali)-[~]
$
```

ii. Open ports discovered:

Commands used: *sudo nmap -p- 10.15.164.0/24*

Host: 10.15.164.74

Port	State	Service
5040	Open	Unknown
5357	Open	WSDAPI
7680	Open	Pando-pub

Host: 10.15.164.85

Port	State	Service
53	Open	DNS
33155	Filtered	Unknown
41959	Filtered	Unknown

```
(kali㉿kali)-[~]
$ sudo nmap -p- 10.15.164.0/24

Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 22:24 -0400
Nmap scan report for 10.15.164.74
Host is up (0.00032s latency).
Not shown: 65532 filtered tcp ports (no-response)
PORT      STATE SERVICE
5040/tcp  open  unknown
5357/tcp  open  wsdapi
7680/tcp  open  pando-pub
MAC Address: 64:D6:9A:B1:FB:CA (Intel Corporate)

Nmap scan report for 10.15.164.85
Host is up (0.0078s latency).
Not shown: 65315 closed tcp ports (reset), 219 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: A6:FF:1F:5C:09:22 (Unknown)
```

iii. Services running on those ports identified:

Host 10.15.164.74:

- Port 5040: Unknown service
- Port 5357: (WSDAPI) -Web Services for Devices
- Port 7680: Windows Delivery Optimization service

Host 10.15.164.85:

- Port 53: Domain Name System (DNS) service
- Other ports could not be fully identified.

```
(kali@kali)-[~]
$ sudo nmap -sV 10.15.164.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 22:31 -0400
Nmap scan report for 10.15.164.74
Host is up (0.00086s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
5357/tcp  open  http      Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 64:D6:9A:B1:FB:CA (Intel Corporate)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for 10.15.164.85
Host is up (0.0090s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
53/tcp    open  domain    dnsmasq 2.51
MAC Address: A6:FF:1F:5C:09:22 (Unknown)

Nmap scan report for 10.15.164.71
Host is up.
All 1000 scanned ports on 10.15.164.71 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (3 hosts up) scanned in 226.98 seconds
```

Number of devices on my local network

A network scan was performed using Nmap to identify active devices on my local network.

The scan discovered 2 active devices:

10.15.164.74

10.15.164.85

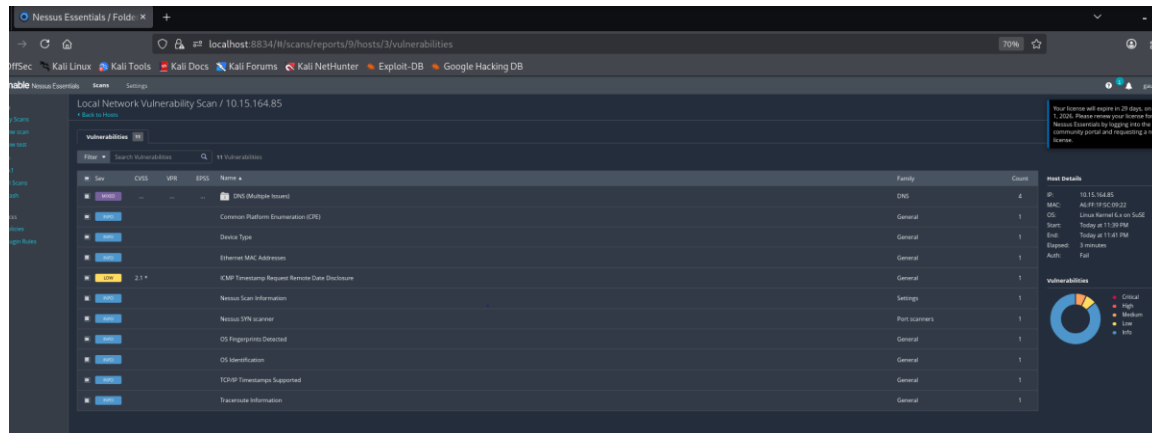
2. Vulnerability Scan on the identified hosts:

a. The correct tool for this job is Nessus as it the scanner used to:

Identify security weaknesses

- Detect missing patches
- Find configuration issues
- Classify vulnerabilities by severity

b. Screenshot



c. Analysing the scan results to identify vulnerabilities.

The severity levels were identified from the Nessus scan dashboard, which groups vulnerabilities according to risk level using severity ratings as the severity levels were determined from the Nessus vulnerability summary which categorises vulnerabilities by severity using CVSS scoring and colour-coded indicators.

